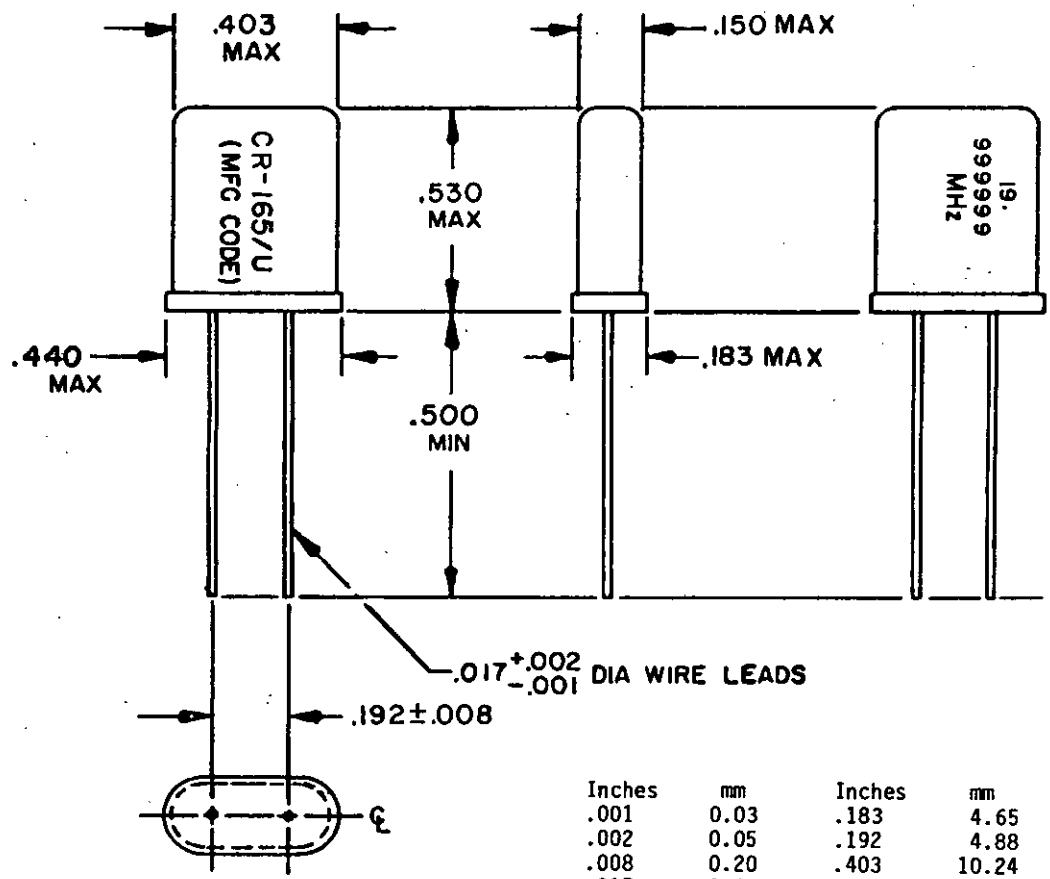
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-165/U

This specification is approved for use within the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 2.4 MHz to 20 MHz; fundamental; noncontrolled; antiresonance.



Inches	mm	Inches	mm
.001	0.03	.183	4.65
.002	0.05	.192	4.88
.008	0.20	.403	10.24
.017	0.43	.435	11.05
.150	3.81	.500	12.70
		.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The type-designation, frequency, and code-designating characters shall be not less than .038 (0.97 mm) in height and not less than .027 (0.69 mm) in width.
4. Corrosion resistant metal plating of the enclosure after resistance welding is permitted.

FIGURE 1. Dimensions and configuration.

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-49/U (MIL-H-10056/36)
Dimensions, marking, and configuration	See figure 1
Frequency range	2.4 MHz to 20 MHz, inclusive
Frequency tolerance (operating temperature range)	±50 ppm
Antiresonance, load capacitance	30.0 pF ±0.5
Equivalent resistance	See table III
Operating temperature range	-55°C to +105°C, inclusive
Mode of oscillation	Fundamental
Test set, calibration values, and rated drive level	See table II
Capacitance, shunt	2.4 MHz to 4.0 MHz--1.7 pF min to 7.0 pF max 4.0 MHz to 6.0 MHz--2.7 pF min to 7.0 pF max 6.0 MHz to 20.0 MHz--3.7 pF min to 7.0 pF max
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 ppm

1/ Method 204 of MIL-STD-202, test condition A.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volt	mW
TS-330/TSM	2.4 to 5.1	50	10	---	5.0 ±1.0
	5.1+ to 7.5	22	15	---	5.0 ±1.0
	7.5+ to 10.0	13	20	---	5.0 ±1.0
TS-683/TSM	10+ to 20	10	---	0.22	5.0 ±1.0

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TABLE III. Equivalent resistance.

Frequency range, inclusive			Maximum resistance
MHz			Ohms
2.4	to	2.60	300
2.61+	to	2.89	250
2.9	to	3.75	180
3.75+	to	4.75	120
4.75+	to	6.0	75
6+	to	7	50
7+	to	10	30
10+	to	20	25

Preparing activity:
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